



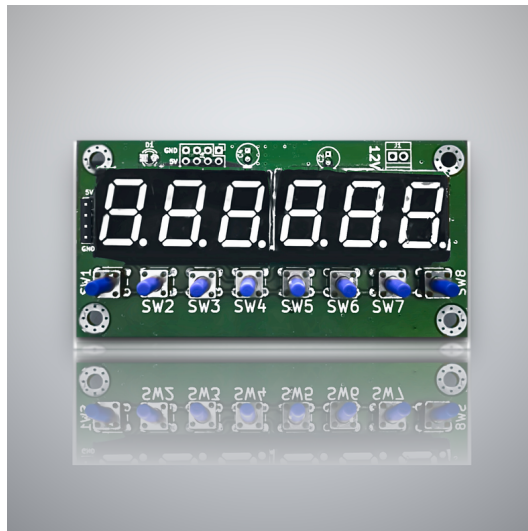
TM1637 6-Digit LED Display & Button Module

Six-digit 7-segment LED display with eight user buttons and a 12 V input, driven by a Titan Micro TM1637 over a two-wire serial interface

The TM1637 6-Digit LED Display & Button Module is a panel-mount numeric readout with eight tactile keys on a single 90 x 45.25 mm board. A host controller drives the whole assembly over two signal wires, CLK and DIO, so a complete display and keypad front end costs four connections.

The module is built for equipment rather than the bench. Power arrives on a two-pin screw terminal through a bridge rectifier and an on-board 5 V regulator, so it runs from an unregulated 12 V control supply, AC or DC. The TM1637 driver sits on the underside, leaving the front face flat for panel mounting behind a cut-out or bezel, and four 3.50 mm holes take standoffs or panel screws.

Micronode supplies the driver source, worked examples and full mechanical data, so the module can be designed into a product rather than bolted on. Boards ship assembled and tested from Pune, India.



Key features

- Six 7-segment digits, each with a decimal point
- Titan Micro TM1637 driver, SOP-20, on the underside
- Two-wire serial interface, CLK and DIO
- Display can be blanked without clearing its contents
- Full driver library and examples on GitHub
- Eight 6 mm tactile buttons along the front edge
- 12 V AC or DC screw terminal
- On-board bridge rectifier and 5 V regulator
- 4-pin host header and 2 x 4 power header
- 90 x 45.25 mm, 4 x 3.50 mm mounting holes

Applications

- Panel meters and process indicators
- Counters, totalisers and batch displays
- Temperature, pressure and flow readouts
- Weighing and dosing terminals
- Machine HMI front panels and set-point entry
- Test, calibration and production-line fixtures
- Motor drive and pump status displays
- Retrofit replacements for obsolete LED panels

1. Display

Driver	Titan Micro TM1637, SOP-20, mounted on the underside of the board
Digits	Six 7-segment digits with decimal points, arranged as two three-digit modules
Module size	37.8 x 19.0 mm each
Interface	Two-wire serial, CLK and DIO, driven directly by a host microcontroller. One driver per pair of pins
Clock rate	Up to 250 kHz on CLK
Blanking	The display can be switched off without losing the digit data



2. Power

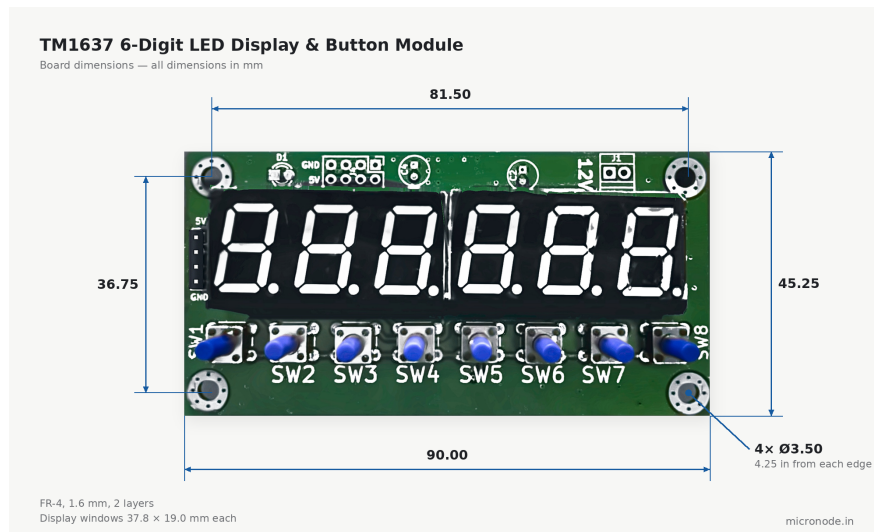
Screw terminal	12 V nominal, AC or DC. A bridge rectifier follows the terminal
Regulator	On-board 5 V regulator in TO-252, feeding the TM1637 and the display
5 V rail	Available on the 2 x 4 header, four positions, and on the display header. The module can be run from 5 V directly when no 12 V supply is applied

3. Controls and indicators

Designator	Type	Function
SW1 to SW8	6 mm tactile switch	User input along the front edge of the board
Power LED	SMD LED	Lit whenever the board is powered, from either the 12 V input or the 5 V rail

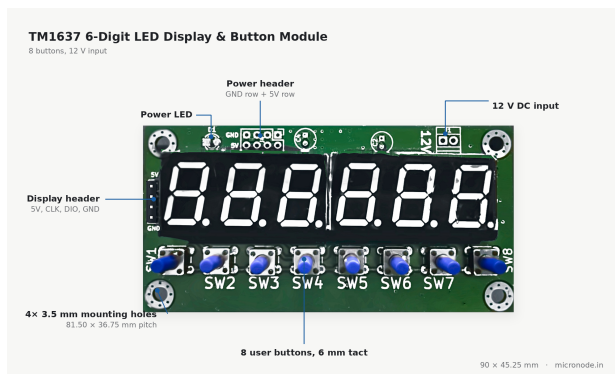
4. Mechanical

Board size	90.00 x 45.25 mm
Thickness	1.6 mm FR-4, two layers, green solder mask, white silkscreen
Mounting	4 x 3.50 mm plated holes on an 81.50 x 36.75 mm pitch, 4.25 mm in from each edge
Component height	8.65 mm above the board face
Connectors	Two-pin screw terminal, one 1 x 4 header, one 2 x 4 header on 2.54 mm pitch



All dimensions in millimetres.

5. Connections



Header	Pin	Signal
Display header	1	5 V rail
	2	CLK
	3	DIO
	4	Ground
Power header	Top row	Ground, four positions
	Bottom row	5 V rail, four positions
Screw terminal	1, 2	12 V AC or DC in

6. Software and drivers

Micronode publishes the full driver source and worked examples for this module on GitHub, together with the digit and segment map for the board. The library is not a minimal example. It covers formatted number output with leading-zero and blanking control, individual digit and segment writes, character and text display, elapsed-time and clock formats using the decimal points, and reading the eight buttons. In most applications one call initialises the module and one call updates the reading, so integration is a short job even for an engineer meeting the part for the first time.

The driver is written in plain C with no dependency on any framework or vendor SDK. Only two pins on the host touch the module, CLK and DIO, and every hardware access in the driver goes through three small functions that the integrator writes for their own device: drive CLK or DIO high or low, read the state of DIO, and wait a few microseconds. Everything above that layer, including the segment tables, number formatting and button decode, compiles unchanged. The same source therefore runs on any microcontroller or microprocessor, from an 8051 or STM32 to a Linux single-board computer.

CLK and DIO are ordinary general purpose pins on the host; no dedicated peripheral or spare serial port is required. DIO is bidirectional, so the host pin must be able to switch between output and input.

7. Ordering information

Order code	Description
MN-TM1637-6D	Display module, fully assembled and tested
MN-TM1637-6D-P	Reduced build, easier to probe and analyse. D1, C2, C4, the 2 x 4 power header and the 12 V screw terminal are not fitted, but the pads remain, so a terminal or flying leads can be added later. The board still runs from 12 V or from 5 V

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Specifications may change without notice; the current revision of this document is published at micronode.in. TM1637 is a product of Titan Micro Electronics. Not qualified for life-support or safety-critical equipment.